

Humaira Mir Prothoma

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🔗 Academic Portfolio 🔗 [linkedin.com/in/humairamirprothoma](https://www.linkedin.com/in/humairamirprothoma)

Education

B.Sc in Computer Science and Engineering, BRAC University
CGPA - 3.44/.00

2019 – 2023

Professional Experience

Training & Content Specialist, Therap (BD)

08/2024 – Present

- Develop technical content, user guides, and release documentation for multiple software modules.
- Collaborate with developers, QA engineers, and designers to improve product usability.
- Utilize AI tools (MaestroQA, Heygen, NotebookLM) and prompt engineering to automate QA workflows and content generation.
- Troubleshoot complex user issues and contribute to API integration documentation.

Research Experience

Undergraduate Thesis — BRAC University *Real-time Unwanted Scenes Detection and Blurring*

- Built a content moderation system using computer vision to safeguard children from inappropriate media.
- Benchmarked YOLOv6 against other detection algorithms for speed and accuracy.
- Applied OpenCV for automated real-time blurring of detected content.

Technical Projects

KineiFelbo.com (Online Shopping Management System)

Tech Stack: HTML, CSS, PHP, Java, MySQL

- Designed and developed a web-based e-commerce platform as part of a team project
- Implemented product catalog, shopping cart, and order management modules
- Strengthened skills in full-stack development and collaborative software engineering

Movie Review Sentiment Analysis (Personal Project)

Tech Stack: Python, Scikit-learn, NLTK, Hugging Face, Gradio, GitHub, Matplotlib, WordCloud

- Built and deployed a sentiment analysis system for movie reviews with Positive, Negative, and Neutral classification
- Implemented TF-IDF + Logistic Regression model (~85% accuracy) with explainability features (top words, word clouds)
- Integrated VADER for neutral detection and optional RoBERTa model for contextual accuracy
- Developed and deployed an interactive Gradio web app on Hugging Face Spaces via GitHub

Kokhon Chai Delivery (Delivery Service Android App)

Tech Stack: Java, Android Studio, Firebase

- Built an Android application for local delivery service, enabling order placement and real-time updates
- Integrated Firebase for authentication and cloud-based data storage
- Gained practical experience in Android-specific concepts and mobile UI design

FocusGRE (Ongoing, Personal Project)

Tech Stack: HTML, CSS, JavaScript

- Developing an interactive platform for GRE Quant & Verbal practice, vocabulary learning, and full-length mock tests
- Implemented dynamic question rendering, progress tracking, and user-friendly UI/UX design
- Focused on building scalable and engaging educational technology tools

Research Interest

I am deeply interested in understanding how people use and experience technology, and I want to design systems that make their lives easier and more meaningful.

- **Natural Language Processing (NLP):** Using language and text data to help technology understand and respond to people more naturally. This can make systems more intuitive, responsive, and user-friendly across different applications.
- **Machine Learning (ML):** Teaching computers to learn from data so they can understand patterns, predict needs, and improve interactions over time. This enables systems to be adaptive, intelligent, and more helpful for all users, including those with accessibility needs.
- **Human-Computer Interaction (HCI):** Creating simple, intuitive, and user-friendly interfaces by studying how people interact with technology in real-life situations.
- **Accessibility:** Building inclusive digital tools that can be used by people with different abilities and needs, so no one is left out of technological progress.
- **Information Retrieval (IR):** Developing smarter systems to organize, search, and retrieve useful information from large amounts of data quickly and accurately.

Technical Skills

Programming Languages

- Java (Android app development — *Kokhon Chai Delivery*)
- JavaScript (Web interactivity — *GRE Practice App*)
- Python (Problem-solving, automation, and machine learning — *Sentiment Analysis Project*)

AI / ML & Automation Tools

- Scikit-learn, NLTK, Hugging Face Transformers, VADER, Gradio (Deployed Sentiment Analysis web app)
- MaestroQA, Heygen, NotebookLM (Used for QA automation and content generation)

Web Development

- HTML, CSS, PHP (Full-stack development — *KineiFelbo.com*)

Databases

- MySQL (Used in e-commerce platform backend design)
- Firebase (Integrated into Android app for cloud storage & authentication)

Tools & Platforms

- Git & GitHub (Version control & deployment), Hugging Face Spaces, Salesforce (Workflow support), VS Code, Android Studio, WordPress

Documentation & Productivity

- Microsoft Office Suite, Google Workspace

Standardized Test Scores

IELTS

7.5 (L: 8.0 | R: 8.0 | W: 6.5 | S: 7.5)

GRE

304 (Q: 159 | V: 145 | AWA: 3.5)